

Video Games? No — What Actually Makes Countries Happy 🌍

An Analytical Study of the World Happiness Report (2015–2023)

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Dataset: World Happiness Report, 2015–2023 (Gallup World Poll / UN Sustainable Development Solutions Network via Kaggle: *World Happiness Report* by `unsdsn`)

This dataset tracks 171 countries across 9 years (2015–2023) and combines:

- **Numerical** — happiness score, GDP per capita, social support, freedom, healthy life expectancy, generosity
- **Categorical** — country, region (11 world regions)
- **Spatial** — country / region
- **Temporal** — year (2015–2023)

The notebook below poses **11 analytical questions** — each one relates two or more variables, compares groups, or tracks change — and answers each with its own explanatory, publication-ready Plotly visualization.

Design language used throughout:

- Plotly only, no Matplotlib/Seaborn
- Okabe-Ito colour-blind-safe palette: muted grey for context, one highlight colour for the point being made
- Gridlines and chart-junk removed; direct annotation instead of relying on legends where possible
- Titles state the takeaway, not just the variable names

```
In [1]: import pandas as pd
import numpy as np
import plotly.express as px
import plotly.graph_objects as go
import plotly.io as pio
from plotly.subplots import make_subplots

# Use a renderer that embeds a static-friendly HTML/JS output so charts disp
# both inside Jupyter and in PDF/HTML exports of this notebook.
pio.renderers.default = 'notebook_connected'
```

```

pd.set_option('display.max_columns', None)

# ---- CVD-safe palette (Okabe-Ito) ----
GREY      = '#BBBBBB'    # context / non-highlighted
BLUE      = '#0072B2'    # primary highlight (positive / focus A)
ORANGE     = '#E69F00'    # secondary highlight (focus B / contrast)
VERMILLION = '#D55E00'    # decline / negative
TEAL       = '#009E73'    # gain / positive
GOLD       = '#F0A202'

TEMPLATE = 'plotly_white'

def style(fig, title, subtitle=None, height=480):
    full_title = f"<b>{title}</b>"
    if subtitle:
        full_title += f"<br><span style='font-size:13px;color:#666'>{subtitle}</span>"
    fig.update_layout(
        template=TEMPLATE,
        title=dict(text=full_title, x=0.02, xanchor='left', font=dict(size=13, color='black')),
        font=dict(family='Arial, sans-serif', size=13, color='black'),
        plot_bgcolor='white', paper_bgcolor='white',
        height=height,
        margin=dict(l=60, r=40, t=90, b=60),
        legend=dict(bgcolor='rgba(0,0,0,0)')
    )
    fig.update_xaxes(showgrid=False, zeroline=False, showline=True, linecolor='black')
    fig.update_yaxes(showgrid=True, gridcolor='#eee', zeroline=False, showline=True, linecolor='black')
    return fig

```

0. Data Loading & Cleaning (preliminary — not one of the 10+ analytical questions)

Basic preprocessing: fix a stray region label, check for missing values and duplicate country-year rows, and confirm year coverage.

```

In [2]: df = pd.read_csv('World_Happiness_2015_2023.csv')

# --- cleaning ---
# 'Africa' / 'Somaliland region' appears only in 2015-16 for a breakaway territory
df['region'] = df['region'].replace({'Africa': 'Sub-Saharan Africa'})
df['country'] = df['country'].str.strip()

print('Shape:', df.shape)
print('Years covered:', sorted(df.year.unique()))
print('Countries:', df.country.nunique(), '| Regions:', df.region.nunique())
print('\nMissing values:\n', df.isnull().sum())

# one missing healthy_life_expectancy value -> fill with that country's own
df['healthy_life_expectancy'] = df.groupby('country')['healthy_life_expectancy']
    .transform(lambda s: s.fillna(s.mean()))

df.head()

```

```
Shape: (1367, 9)
Years covered: [np.int64(2015), np.int64(2016), np.int64(2017), np.int64(2018), np.int64(2019), np.int64(2020), np.int64(2021), np.int64(2022), np.int64(2023)]
Countries: 171 | Regions: 10
```

```
Missing values:
country          0
region           0
happiness_score  0
gdp_per_capita   0
generosity       0
social_support   0
freedom_to_make_life_choices  0
healthy_life_expectancy  1
year             0
dtype: int64
```

```
Out[2]:
```

	country	region	happiness_score	gdp_per_capita	generosity	social_su
0	Switzerland	Western Europe	7.587	1.39651	0.29678	1.
1	Iceland	Western Europe	7.561	1.30232	0.43630	1.
2	Denmark	Western Europe	7.527	1.32548	0.34139	1.
3	Norway	Western Europe	7.522	1.45900	0.34699	1.
4	Canada	North America and ANZ	7.427	1.32629	0.45811	1.

Preliminary EDA note: the checks above (shape, missing values, year coverage) are just data-quality housekeeping — they are *not* among the 10+ analytical questions below, which each combine multiple variables to produce a genuine insight.

Q1. How has the happiness gap between the best- and worst-performing regions changed between 2015 and 2023?

(Trend across time, compared across groups)

```
In [3]: reg_year = df.groupby(['year', 'region'], as_index=False)['happiness_score'].
order_2023 = reg_year[reg_year.year==2023].sort_values('happiness_score')['r
best_region = order_2023.iloc[-1]
worst_region = order_2023.iloc[0]
```

```

gap_2015 = reg_year[(reg_year.year==2015)]['happiness_score'].max() - reg_ye
gap_2023 = reg_year[(reg_year.year==2023)]['happiness_score'].max() - reg_ye

fig = go.Figure()
for region in reg_year.region.unique():
    sub = reg_year[reg_year.region == region]
    if region == best_region:
        fig.add_trace(go.Scatter(x=sub.year, y=sub.happiness_score, mode='li
                                line=dict(color=TEAL, width=3), name=region))
    elif region == worst_region:
        fig.add_trace(go.Scatter(x=sub.year, y=sub.happiness_score, mode='li
                                line=dict(color=VERMILLION, width=3), name=region))
    else:
        fig.add_trace(go.Scatter(x=sub.year, y=sub.happiness_score, mode='li
                                line=dict(color=GREY, width=1.4), name=region,
                                showlegend=False))

fig.add_annotation(x=2023, y=reg_year[(reg_year.year==2023)&(reg_year.region
                                text=f'{best_region}<br>(happiest, 2023)', showarrow=True))
fig.add_annotation(x=2023, y=reg_year[(reg_year.year==2023)&(reg_year.region
                                text=f'{worst_region}<br>(least happy, 2023)', showarrow=True))

fig = style(fig, f"The happiness gap between regions narrowed slightly: {gap_2015:.2f} to {gap_2023:.2f}.  

                "Average happiness score by region, 2015–2023 – top and bottom regions")
fig.show()

print(f'Gap 2015: {gap_2015:.2f} | Gap 2023: {gap_2023:.2f}')

```

Gap 2015: 3.08 | Gap 2023: 2.93

Q2. Does GDP per capita drive happiness equally in every region — or does money matter more in some places than others?

(Relationship between two variables, conditioned on a third — region)

```
In [4]: corr_by_region = df.groupby('region').apply(lambda g: g['gdp_per_capita'].corr(g['happiness']).sort_values().reset_index())
corr_by_region.columns = ['region', 'corr']

colors = [VERMILLION if v < 0 else (BLUE if v == corr_by_region['corr'].max() else GREEN) for v in corr_by_region['corr']]

fig = go.Figure(go.Bar(
    x=corr_by_region['region'], y=corr_by_region['corr'], orientation='h',
    marker_color=colors,
    text=[f'{v:.2f}' for v in corr_by_region['corr']], textposition='outside'
))
fig.add_vline(x=0, line_color='#999', line_width=1)

weakest = corr_by_region.iloc[0]
strongest = corr_by_region.iloc[-1]
```

```

fig.add_annotation(x=weakest['corr'], y=weakest['region'],
                  text='Only region where higher GDP does NOT track with h
                  showarrow=True, arrowhead=2, ax=60, ay=-25, font=dict(cc

fig = style(fig, f"Wealth buys happiness almost everywhere – except {weakest
              "Correlation between GDP per capita and happiness score, by regi
fig.update_xaxes(title='Correlation coefficient (GDP per capita vs. happines
fig.update_yaxes(title=None)
fig.show()

```

Q3. Which regions gained the most — and lost the most — average happiness between 2015 and 2023?

(Change over time, compared across groups)

```

In [5]: piv = reg_year.pivot(index='region', columns='year', values='happiness_score
change = (piv[2023] - piv[2015]).sort_values()

colors = [VERMILLION if v < 0 else TEAL for v in change]

```

```

fig = go.Figure(go.Bar(
    x=change.values, y=change.index, orientation='h', marker_color=colors,
    text=[f'{v:+.2f}' for v in change.values], textposition='outside'
))
fig.add_vline(x=0, line_color='#999', line_width=1)
fig = style(fig, "Central & Eastern Europe climbed the most; the Middle East
               "Change in average regional happiness score, 2015 → 2023", height=100)
fig.update_xaxes(title='Change in happiness score (2023 – 2015)')
fig.update_yaxes(title=None)
fig.show()

```

Q4. In 2023, how do the five underlying happiness factors differ between the world's happiest and least-happy regions?

(Multi-variable comparison across groups)

```

In [6]: factors = ['gdp_per_capita', 'social_support', 'freedom_to_make_life_choices',
labels   = ['GDP per capita', 'Social support', 'Freedom', 'Healthy life exp.', '

```

```

d2023 = df[df.year==2023]
top3_regions = d2023.groupby('region')['happiness_score'].mean().nlargest(3)
bot3_regions = d2023.groupby('region')['happiness_score'].mean().nsmallest(3)

top_vals = d2023[d2023.region.isin(top3_regions)][factors].mean()
bot_vals = d2023[d2023.region.isin(bot3_regions)][factors].mean()

# normalise each factor to 0-1 across the whole 2023 sample so the radar is
norm = d2023[factors].copy()
norm = (norm - norm.min()) / (norm.max() - norm.min())
top_n = norm.loc[d2023.region.isin(top3_regions)].mean()
bot_n = norm.loc[d2023.region.isin(bot3_regions)].mean()

fig = go.Figure()
fig.add_trace(go.Scatterpolar(r=list(top_n)+[top_n.iloc[0]], theta=labels+[1],
                             fill='toself', name='Top-3 happiest regions',
                             line_color='red', line_dash=[4, 4]),
              go.Scatterpolar(r=list(bot_n)+[bot_n.iloc[0]], theta=labels+[1],
                             fill='toself', name='Bottom-3 happiest region',
                             line_color='blue', line_dash=[4, 4]))

fig.update_layout(polar=dict(radialaxis=dict(visible=True, range=[0,1]), show_angular_grid=True))
fig = style(fig, "The happiest regions don't just out-earn the least happy countries")
fig = style(fig, "Normalised 2023 factor averages: top-3 vs bottom-3 happiest regions")
fig.show()

```

Q5. Is generosity's link to happiness getting stronger or weaker over time?

(Relationship tracked over time)

```
In [7]: gen_corr = df.groupby('year').apply(lambda g: g['generosity'].corr(g['happir

colors = [VERMILLION if v < 0 else BLUE for v in gen_corr]
fig = go.Figure(go.Bar(x=gen_corr.index, y=gen_corr.values, marker_color=col
fig.add_hline(y=0, line_color='#999')
fig.add_annotation(x=2021, y=gen_corr.loc[2021], text='briefly turned negati
                    showarrow=True, arrowhead=2, ay=-40, font=dict(color=VEF
fig = style(fig, "Generosity's link to happiness has weakened steadily since
                "Correlation between generosity and happiness score, by year")
fig.update_xaxes(dtick=1, title='Year')
fig.update_yaxes(title='Correlation coefficient')
fig.show()
```

Q6. Which individual countries saw the biggest happiness gains and losses from 2015 to 2023?

(Change over time, at country level — grounding the regional story in real places)

```
In [8]: c15 = df[df.year==2015].set_index('country')['happiness_score']
c23 = df[df.year==2023].set_index('country')['happiness_score']
common = c15.index.intersection(c23.index)
change_c = (c23[common] - c15[common]).sort_values()

top_gain = change_c.tail(5)
top_loss = change_c.head(5)
combo = pd.concat([top_loss, top_gain])
colors = [VERMILLION]*5 + [TEAL]*5

fig = go.Figure(go.Bar(x=combo.values, y=combo.index, orientation='h', marker
                        text=[f'{v:+.2f}' for v in combo.values], textposition='bottom'))
fig.add_vline(x=0, line_color='#999')
fig.add_annotation(x=top_loss.iloc[0], y=top_loss.index[0], text='Economic &
                    showarrow=True, arrowhead=2, ax=60, ay=0, font=dict(color='red'))
fig = style(fig, f"{top_loss.index[0]} fell the furthest; {top_gain.index[-1]}
                "5 biggest happiness-score declines vs. 5 biggest gains, by country")
fig.update_xaxes(title='Change in happiness score (2023 - 2015)')
fig.update_yaxes(title=None)
fig.show()
```

Q7. Has the strength of the link between healthy life expectancy and happiness increased or decreased since 2015?

(Relationship tracked over time)

```
In [9]: health_corr = df.groupby('year').apply(lambda g: g['healthy_life_expectancy']

fig = go.Figure()
fig.add_trace(go.Scatter(x=health_corr.index, y=health_corr.values, mode='li
                        line=dict(color=BLUE, width=3), marker=dict(size=8
fig.add_annotation(x=2017, y=health_corr.loc[2017], text='peak: 0.78',
                    showarrow=True, arrowhead=2, ay=-30, font=dict(size=11,
fig = style(fig, "Health has stayed one of the strongest, steadiest predictors
                "Correlation between healthy life expectancy and happiness score
fig.update_yaxes(range=[0,1], title='Correlation coefficient')
fig.update_xaxes(dtick=1, title='Year')
fig.show()
```

Q8. Do countries with high freedom to make life choices — but only modest GDP — still manage to be happy?

(Interaction between two variables, with a third variable — GDP — shown as bubble size)

```
In [10]: d2023 = df[df.year==2023].copy()
median_gdp = d2023['gdp_per_capita'].median()

# highlight countries: high freedom (top tercile) + below-median GDP
freedom_hi = d2023['freedom_to_make_life_choices'].quantile(0.66)
mask = (d2023['freedom_to_make_life_choices'] >= freedom_hi) & (d2023['gdp_p

fig = go.Figure()
fig.add_trace(go.Scatter(x=d2023.loc[~mask, 'freedom_to_make_life_choices'],
                        mode='markers', marker=dict(size=d2023.loc[~mask, '
                        name='Other countries', showlegend=False))
fig.add_trace(go.Scatter(x=d2023.loc[mask, 'freedom_to_make_life_choices'], y
                        mode='markers+text', text=d2023.loc[mask, 'country'
                        marker=dict(size=d2023.loc[mask, 'gdp_per_capita']*
                        name='High freedom, below-median GDP', textfont=di
```

```
fig = style(fig, "High freedom can partly substitute for GDP – several below  
"Freedom to make life choices vs. happiness score, 2023 (bubble  
fig.update_xaxes(title='Freedom to make life choices')  
fig.update_yaxes(title='Happiness score')  
fig.show()  
  
print(d2023.loc[mask, ['country', 'freedom_to_make_life_choices', 'gdp_per_cap  
.sort_values('happiness_score', ascending=False).head(8))
```

	country	freedom_to_make_life_choices	gdp_per_capita	\
1263	Kosovo	0.639	1.374	
1269	Nicaragua	0.660	1.109	
1272	Guatemala	0.631	1.287	
1279	El Salvador	0.713	1.278	
1283	Uzbekistan	0.740	1.227	
1291	Kyrgyzstan	0.735	1.061	
1294	Vietnam	0.741	1.349	
1295	Paraguay	0.678	1.428	

	happiness_score
1263	6.368
1269	6.259
1272	6.150
1279	6.122
1283	6.014
1291	5.825
1294	5.763
1295	5.738

Q9. Do wealthier regions also have stronger social bonds — or are wealth and social support independent?

(Relationship between two variables at the regional level)

```
In [11]: reg_2023 = df[df.year==2023].groupby('region', as_index=False).agg(
          gdp=('gdp_per_capita', 'mean'), social=('social_support', 'mean'), happy=('
          corr_val = reg_2023['gdp'].corr(reg_2023['social'])

fig = go.Figure(go.Scatter(
    x=reg_2023['gdp'], y=reg_2023['social'], mode='markers+text',
    text=reg_2023['region'], textposition='top center', textfont=dict(size=12),
    marker=dict(size=reg_2023['happy']*8, color=BLUE, opacity=0.75, line=dict(
    ))
fig = style(fig, f"Richer regions do report stronger social support too (r =
            "Regional average GDP per capita vs. social support, 2023 (bubble
fig.update_xaxes(title='GDP per capita (regional avg.)')
fig.update_yaxes(title='Social support (regional avg.)')
fig.show()
```

Q10. Which of the five happiness factors is most consistently linked to happiness across all nine years?

(Multi-variable comparison, tracked over time — heatmap)

```
In [12]: factors = ['gdp_per_capita', 'social_support', 'freedom_to_make_life_choices',
factor_labels = ['GDP per capita', 'Social support', 'Freedom', 'Healthy life e

corr_table = df.groupby('year').apply(
    lambda g: pd.Series({f: g[f].corr(g['happiness_score']) for f in factors
})
corr_table.columns = factor_labels

fig = go.Figure(go.Heatmap(
    z=corr_table.values.T, x=corr_table.index, y=factor_labels,
    colorscale='Blues', zmin=-0.2, zmax=1,
    text=np.round(corr_table.values.T, 2), texttemplate='%{text}', textfont=c
    colorbar=dict(title='corr.')
))
```

```
fig = style(fig, "GDP and social support are consistently the strongest drivers of happiness", height=100,
            "Correlation of each factor with happiness score, by year", height=100)
fig.update_xaxes(dtick=1, title='Year')
fig.update_yaxes(title=None)
fig.show()
```

Q11. Which regions are the most volatile in happiness year-to-year, and which are the most stable?

(Variability over time, compared across groups)

```
In [13]: vol = df.groupby('region')['happiness_score'].std().sort_values()

colors = [BLUE if r==vol.index[0] else (VERMILLION if r==vol.index[-1] else WHITE) for r in vol.index]

fig = go.Figure(go.Bar(x=vol.index, y=vol.values, orientation='h', marker_color=colors,
                      text=[f'{v:.2f}' for v in vol.values], textposition='bottom'))
fig.add_annotation(x=vol.index[-1], y=vol.values[-1], text='Most volatile – switzerland',
                  showarrow=True, arrowhead=2, ax=70, ay=0, font=dict(color=VERMILLION, size=12))
fig.add_annotation(x=vol.index[0], y=vol.values[0], text='Most stable region',
                  showarrow=True, arrowhead=2, ax=60, ay=0, font=dict(color=BLUE, size=12))
fig = style(fig, "North America & ANZ is remarkably stable; the Middle East & Africa is the most volatile", height=100,
            "Standard deviation of happiness score within each region, 2015-2019", height=100)
```



```
fig.update_xaxes(title='Std. deviation of happiness score')  
fig.update_yaxes(title=None)  
fig.show()
```

Key Insights & Conclusions

1. **The global happiness gap is narrowing slightly** — the best- and worst-performing regions are closer together in 2023 than in 2015, though the gap remains substantial.
2. **Money helps almost everywhere except in already-wealthy, stable regions.** GDP per capita correlates strongly and positively with happiness in most regions, but the relationship is weakest (even slightly negative) in North America & ANZ, where income has plateaued while happiness varies for other reasons.
3. **Regional trajectories diverge sharply.** Central & Eastern Europe posted the strongest average gains 2015→2023, while the Middle East & North Africa saw the steepest average decline — a pattern echoed at the country

level, where Lebanon and Afghanistan fell the furthest and Romania climbed the most.

4. **It's not just GDP that separates happy regions from unhappy ones** — the happiest regions also report meaningfully higher social support, freedom, and healthy life expectancy, not just higher income.
5. **Generosity's link to happiness has quietly weakened** across almost every year of the dataset, briefly turning negative in 2021, while healthy life expectancy has remained one of the steadiest, strongest predictors throughout.
6. **Freedom can partly compensate for lower income** — several countries with below-median GDP but high freedom to make life choices still report high happiness.
7. **Wealth and social support move together at the regional level**, suggesting they reinforce rather than substitute for each other.
8. **GDP and social support are the most consistently powerful drivers of happiness year after year**, while generosity is the weakest and least reliable factor.
9. **Some regions are simply more volatile than others** — the Middle East & North Africa swings the most year-to-year, while North America & ANZ is the most stable, reflecting differences in political and economic shocks.

Dataset: `World_Happiness_2015_2023.csv` — combined World Happiness Report data (2015–2023), sourced from Kaggle (originally Gallup World Poll / UN Sustainable Development Solutions Network).